MBAI 5600G – Applied Integrative Analytics Capstone Project

Milestone 1 – Capstone Project Proposal

Multimodal Financial Crisis Prediction: Integrating Market Signals with LLM-Based Financial News Sentiment Analysis

Student Group Name: Group 13

Team Members: Jeya Surya Balaji, Keertan Jingeshkumar Patel

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Instructor: Prof Amin Ibrahim

1. Executive Summary

There are hidden states of financial markets periods of stable growth, increased volatility, and actual crisis which are sometimes not apparent until substantial losses occur. Although traditional quantitative models are only based on price-based signals, a growing number of studies indicates that language signals in financial news may have an impact on the price movements during structural stress.

This is a capstone project that builds a multimodal financial crisis prediction model by integrating two independent analytical processes: (1) a quantitative pipeline with Hidden Markov Models (HMM), ARMA-GARCH volatility modelling on the market data of the S&P 500, and (2) a natural language processing (NLP) pipeline featuring FinBERT, a large language model trained on financial domain data, to extract sentiment signals from financial news headlines. The main research question is: Can the sentiment signals derived from text precede the price signals derived from the data for regime change, and by how many days?

The framework is tested against three known crisis events including the Global Financial Crisis in 2008, the COVID-19 market crash (2020) and the period of volatility during the inflation crisis (2022). Using explainability analysis with SHAP, the signals that trigger each regime transition (price or sentiment) are identified. It is completely done in Python and Google Colab using public dataset that is less than 1GB of size and is implemented within 12 weeks of academic course.

2. Problem Statement & Project Objectives

2.1 Formal Problem Statement

Develop a multimodal financial regime detection and early warning system that integrates Hidden Markov Models, ARMA-GARCH volatility modeling, and FinBERT-based financial news sentiment analysis to detect market regime transitions in S&P 500 equity markets. The system will quantify whether sentiment signals derived from financial news headlines lead or lag price-based regime signals, providing an explainable early warning framework validated against historical financial crises, within a 12-week academic timeline using Python-based tools and publicly available datasets under 1 GB.

2.2 SMART Analysis

SMART Element	Project Alignment
Specific	Detect financial market regimes and quantify lead-lag relationships between price signals (HMM + GARCH) and sentiment signals (FinBERT) in S&P 500 data.
Measurable	Evaluate using HMM log-likelihood and BIC/AIC for model selection; FSI correlation with crisis dates; sentiment signal lead time (days) ahead of price signal during 3 validated crises; SHAP feature importance scores.
Achievable	Uses pre-trained FinBERT (no GPU training required), publicly available datasets, and lightweight statistical models — all compatible with Google Colab free tier.
Relevant	Addresses a real gap in quantitative finance: whether unstructured text data provides leading indicators of systemic market stress beyond what price data alone can signal.

Time-bound	Completed within 12-week capstone timeline, with Person A and Person B working on independent parallel pipelines that integrate in Milestones 5–6.

2.3 Key Objectives

- Collect and preprocess S&P 500 historical price data and CBOE VIX index data from Yahoo Finance
- Collect dated financial news headline datasets aligned with the S&P 500 time series
- Engineer quantitative features: log returns, rolling volatility, maximum drawdown, volatility spikes
- Implement ARMA-GARCH model for conditional volatility estimation
- Implement a 3-state Hidden Markov Model (stable, volatile, crisis) using quantitative features
- Construct a Financial Stress Index (FSI) combining volatility, return shocks, and drawdown intensity
- Apply pre-trained FinBERT model to generate daily sentiment scores from financial news headlines
- Construct sentiment-based stress indicators (fear index, panic signal) from FinBERT outputs
- Perform lead-lag analysis: determine whether sentiment signals precede price signals in regime transitions
- Build a multimodal fusion model combining quantitative and NLP signals for regime prediction
- Apply SHAP explainability to identify which features drive each regime classification
- Validate all findings against three historical crises: 2008 GFC, COVID-19 (2020), and 2022 inflation shock
- Visualize regime timelines, sentiment trends, stress indices, and SHAP feature importance

2.4 Base Research Article

Selected Paper:

[1] L. Wang, S. An, Z. Dong, X. Dong, and J. Li, "Early warning of regime switching in a financial time series: A heteroskedastic network model," PLOS ONE, vol. 20, **no. 10, Art. no. e0333734,** Oct. 2025. DOI: 10.1371/journal.pone.0333734

[2] D. Araci, "FinBERT: Financial sentiment analysis with pre-trained language models," arXiv preprint arXiv:1908.10063, **Aug. 2019. [Online]. Available: https://arxiv.org/abs/1908.10063**

Why these papers were selected:

Wang et al. (2025) provides the peer-reviewed quantitative foundation for the regime detection pipeline, directly informing the HMM and GARCH methodology used by Person A. FinBERT (Araci) is the pre-trained model used for sentiment extraction, with extensive validation in the financial NLP literature through 2025.

How this project extends the existing research:

The prior literature treats quantitative models and NLP sentiment models as separate research streams. This project's novel contribution is the systematic integration of both into a single multimodal framework, with an explicit empirical test of whether text signals lead price signals in regime transitions — a question not answered definitively by any single prior study. The addition of SHAP-based explainability to the regime detection process further extends Wang et al. (2025), which does not address model interpretability.

3. Project Scope

3.1 Included Components

Category	Components
Algorithms & Models	Hidden Markov Model (3-state), ARMA-GARCH, FinBERT (pre-trained), multimodal fusion model, SHAP explainability

Datasets	S&P 500 daily OHLCV (Yahoo Finance), CBOE VIX Index, financial news headlines (Kaggle / public sources), FRED macroeconomic data
Analytical Techniques	Regime detection, volatility modeling, sentiment extraction, lead-lag analysis, Financial Stress Index construction, crisis event validation
Evaluation Metrics	HMM: log-likelihood, BIC/AIC; GARCH: standardized residual diagnostics; FSI: correlation with crisis onset dates; Sentiment lead time: days ahead of price signal; SHAP: feature importance scores per regime
Visualization	Regime timeline overlays, sentiment trend charts, FSI evolution, SHAP bar plots, lead-lag scatter plots
Programming Stack	Python 3.10+, Google Colab; Libraries: hmmlearn, arch, transformers (HuggingFace), pandas, numpy, matplotlib, seaborn, shap, yfinance, fredapi

3.2 Excluded Components

- Real-time data ingestion or live trading systems
- Automated trade execution or brokerage API integration
- Deep learning model training from scratch (LSTM, Transformer training — only pre-trained FinBERT inference)
- High-frequency or intraday trading data
- Production deployment, REST APIs, or web application interfaces
- Large-scale distributed computing or GPU infrastructure
- Non-equity asset classes beyond S&P 500 and VIX as validation inputs

4. Data Sourcing Plan

Dataset	Source	Used By	Size Estimate
S&P 500 daily OHLCV (1990–2024)	Yahoo Finance via yfinance Python library	Person A	< 5 MB
CBOE VIX Index daily (1990–2024)	Yahoo Finance via yfinance Python library	Person A	< 2 MB
FRED macroeconomic indicators	Federal Reserve FRED API	Person A	< 10 MB
Financial news headlines with dates	Kaggle: Financial News and Stock Price Integration Dataset	Person B	< 200 MB
Financial news sentiment (backup)	Kaggle: Daily Financial News for 6000+ Stocks	Person B	< 300 MB
FinBERT pre-trained weights	HuggingFace Hub (ProsusAI/finbert)	Person B	~440 MB (loaded at runtime)

All datasets remain well under 1 GB in total storage. FinBERT weights are loaded at runtime from HuggingFace and do not require persistent storage. Backup sources include Alpha Vantage API (quantitative data) and additional Kaggle financial time-series datasets if primary sources become unavailable.

5. Risk Assessment & Mitigation Plan

Risk	Likelihood	Mitigation Strategy

News headline dates do not align with S&P 500 trading dates	Medium	Align both time series to trading days; forward-fill missing news dates with neutral sentiment score (0.0)
Fin-BERT inference is slow on Colab CPU	Medium	Batch headlines in chunks of 32; use Colab GPU runtime if available; cache sentiment scores to CSV after first run
HMM regime separation is weak (states overlap)	Medium	Tune number of states (2–4); apply feature scaling; use multiple random initializations and select highest log-likelihood
Poor-quality or sparse news data during key crisis periods	Low–Medium	Use two Kaggle backup datasets; supplement with FRED financial commentary data; validate coverage before Milestone 3
GARCH model fails to converge	Low	Use GARCH(1,1) as default; fall back to rolling standard deviation if convergence fails; document assumption
Lead-lag result is null (no signal lead)	Low	A null result is scientifically valid and reportable; document as a finding that price and sentiment signals are contemporaneous
Scope creep between two pipelines	Medium	Define integration interface (shared CSV of daily dates, regime labels, sentiment scores) by Milestone 3; no scope additions after Milestone 4
Computational limits on Google Colab	Low	All models are lightweight statistical or inference-only; no GPU training; expected runtime under 30 minutes per milestone

6. References

[1] L. Wang, S. An, Z. Dong, X. Dong, and J. Li, "Early warning of regime switching in a financial time series: A heteroskedastic network model," *PLOS ONE*, vol. 20, no. 10, Art. no. e0333734, Oct. 2025. DOI: 10.1371/journal.pone.0333734

[2] D. Araci, "FinBERT: Financial sentiment analysis with pre-trained language models," *arXiv preprint* arXiv:1908.10063, Aug. 2019. [Online]. Available: https://arxiv.org/abs/1908.10063

[3] T. Bollerslev, "Generalized autoregressive conditional heteroskedasticity," *Journal of Econometrics*, vol. 31, no. 3, pp. 307–327, Apr. 1986. DOI: 10.1016/0304-4076(86)90063-1

[4] J. D. Hamilton, "A new approach to the economic analysis of nonstationary time series and the business cycle," *Econometrica*, vol. 57, no. 2, pp. 357–384, Mar. 1989. DOI: 10.2307/1912559

[5] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, Long Beach, CA, USA, vol. 30, 2017, pp. 4765–4774.